



MG-1P Damage Assessment Procedures



Visual Inspection



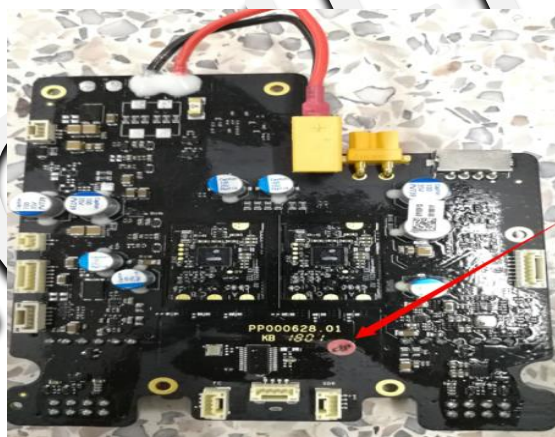
1. Remove all the propellers with the 2.5 hexagonal socket screwdriver and check whether the propellers are damaged or broken.
2. Rotate the motors and check whether they are misshapen, scratched, or obstructed.
3. Pull the rotors vertically and check whether the gap between the stator and the rotor is normal.
4. Check whether the arms cracked. Rotate the arms and check whether they are misshapen.
5. Remove the screws securing the upper shell and remove the upper shell.
6. Check whether the upper shell and lower shell are misshapen or damaged and check whether the screw bases in the airframe are damaged.

Visual Inspection

10. Check whether the carbon tubes of the left and right landing gears are cracked or damaged.
11. Check whether the Spray Tank is damaged or cracked.
12. Check whether the Spray Tank is misshapen or damaged.
13. Check whether the FPV lens is dirty. If yes, clean it with a clean cloth. Check whether the FPV lens is broken or scratched.



11. Check whether the liquid contact indicator on each PCBA has turned red.
12. Check whether the radar's external shell is damaged. If yes, disassemble the shell and check whether the grating disc is misshapen. Rotate the radar and check whether it is obstructed.



Visual Inspection

- Check whether the remote controller is damaged.
- Check whether the battery compartment (battery's pin) and network card slot are damaged.
- Check whether the display's bracket is misshapen or tilted.
- Check whether the screen is broken.
- Disconnect the remote controller's antenna connector and check whether the connector is damaged.
- Push the control stick and check whether the stick is misshapen, tilted or obstructed.



- Check whether the remote controller's antenna is broken.
Solution: If yes, replace the antenna.
- Check whether the upper and lower shells are broken.
Solution: If yes, replace the shell.
- Check whether the remote controller's buttons can spring back after being pressed.
Solution: If not, replace the button and button board.
- Check whether the remote controller's USB port is broken or misshapen.
Solution: If yes, replace the remote controller's small board.
- Check whether the remote controller's display is damaged.
Solution: If yes, replace the screen.

Functional Inspection Before powering on

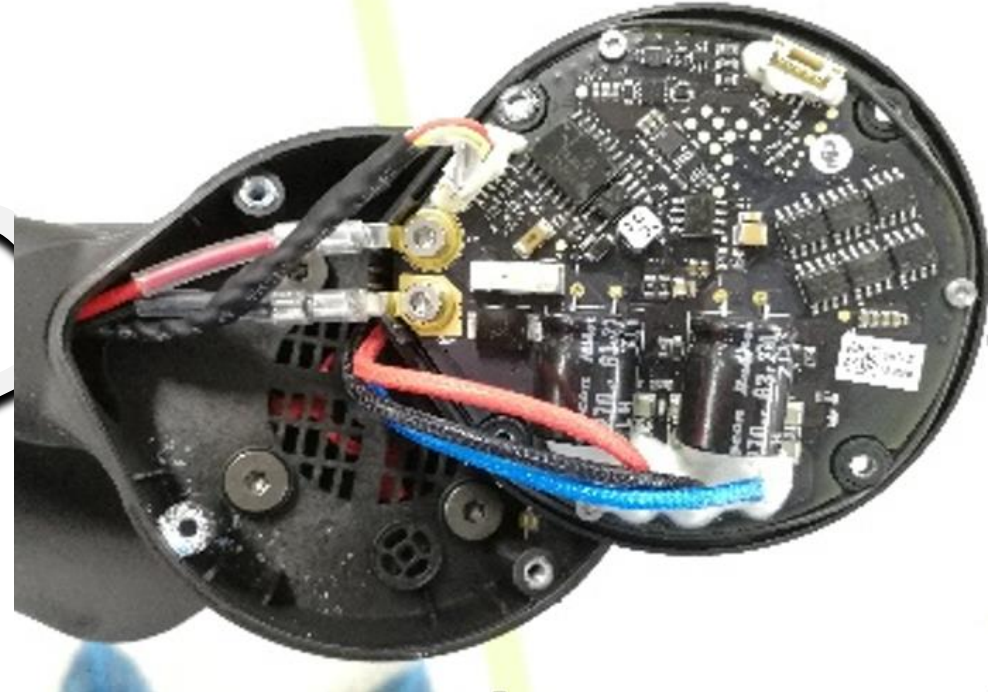


Before powering on

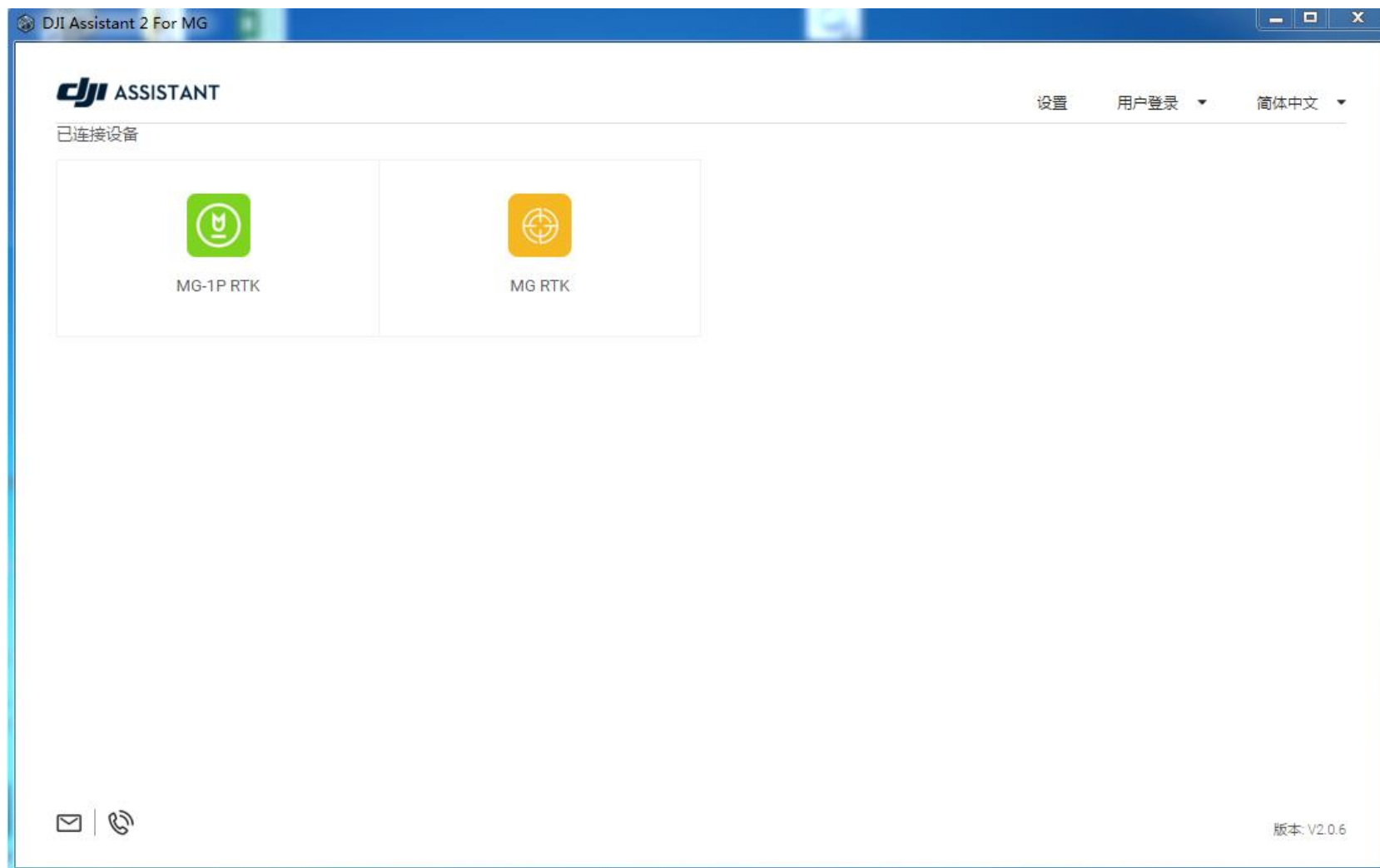
- Open the cover of the defective ESC and check whether the power cables are connected to correct anode and cathode.
- Check whether the socket and cable are damaged or misshapen.
- Check whether the ESC board has been burnt.

If not, power on the product.

1. If the ESC board has been burnt, replace the ESC board.
2. If the cable is damaged, replace the cable.
3. If the drone crashed severely, but the board is not short-circuited, perform the powering on test. If the power is short-circuited, troubleshoot the short-circuiting issue first before performing the test.



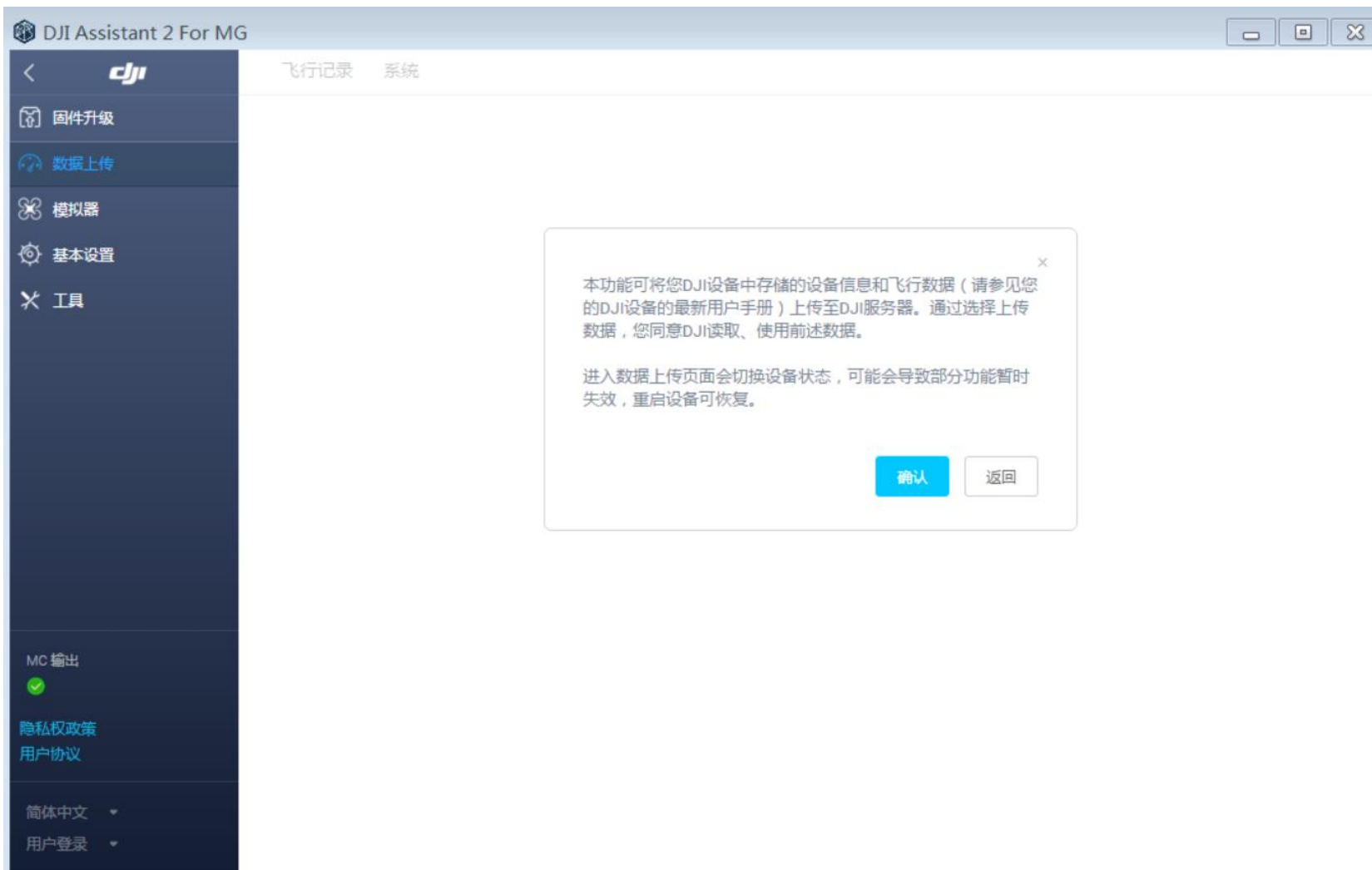
Functional Inspection



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DJI Assistant 2 for MG –
Aircraft/Remote
Controller Firmware
Update – Basic Settings

Functional Inspection

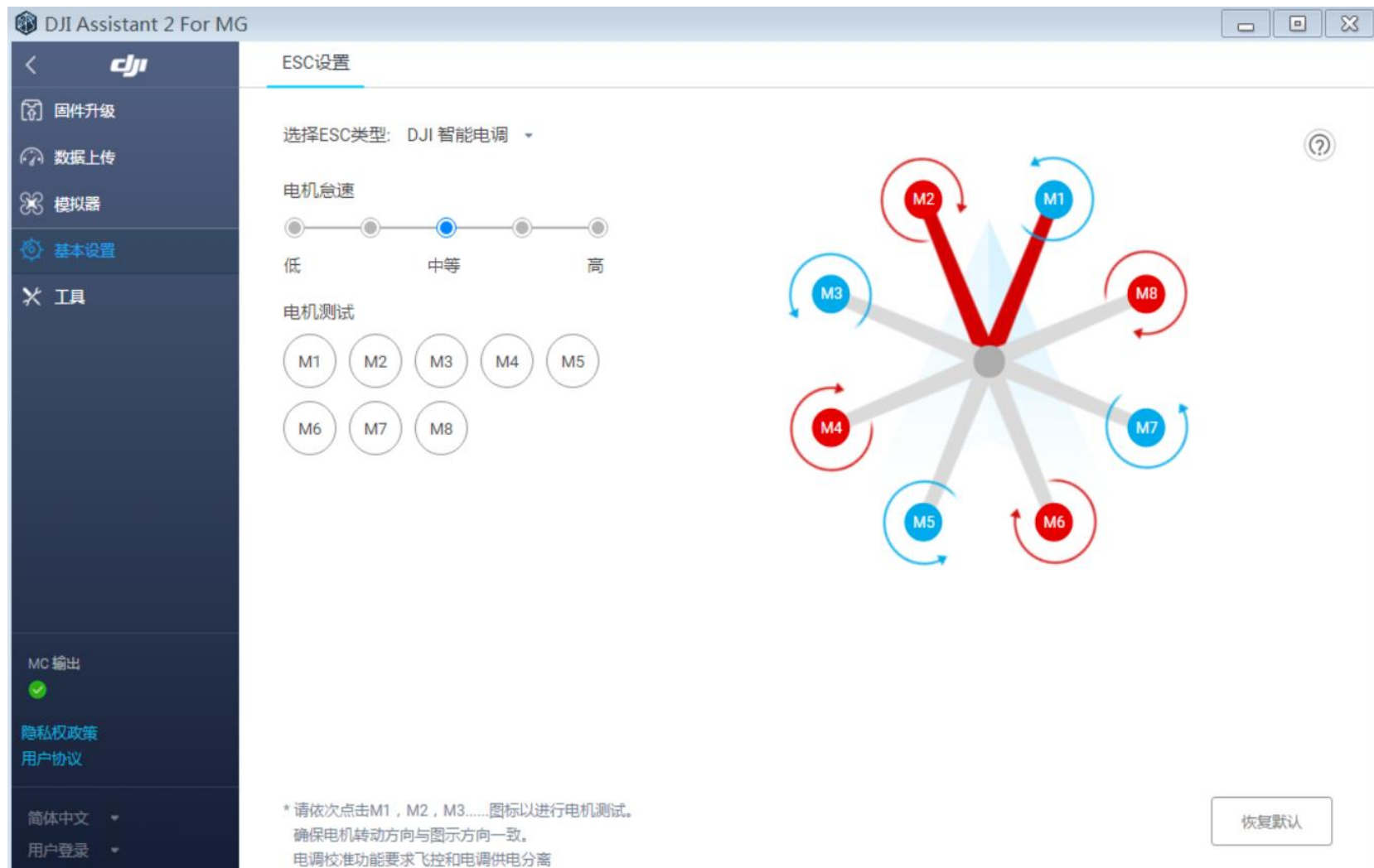


If the drone's issue cannot be determined with visual inspection, export the flight records for analysis.

Click Flight Data and Confirm, open my computer, local disk, and FLG disk, then copy and paste the flight logs.

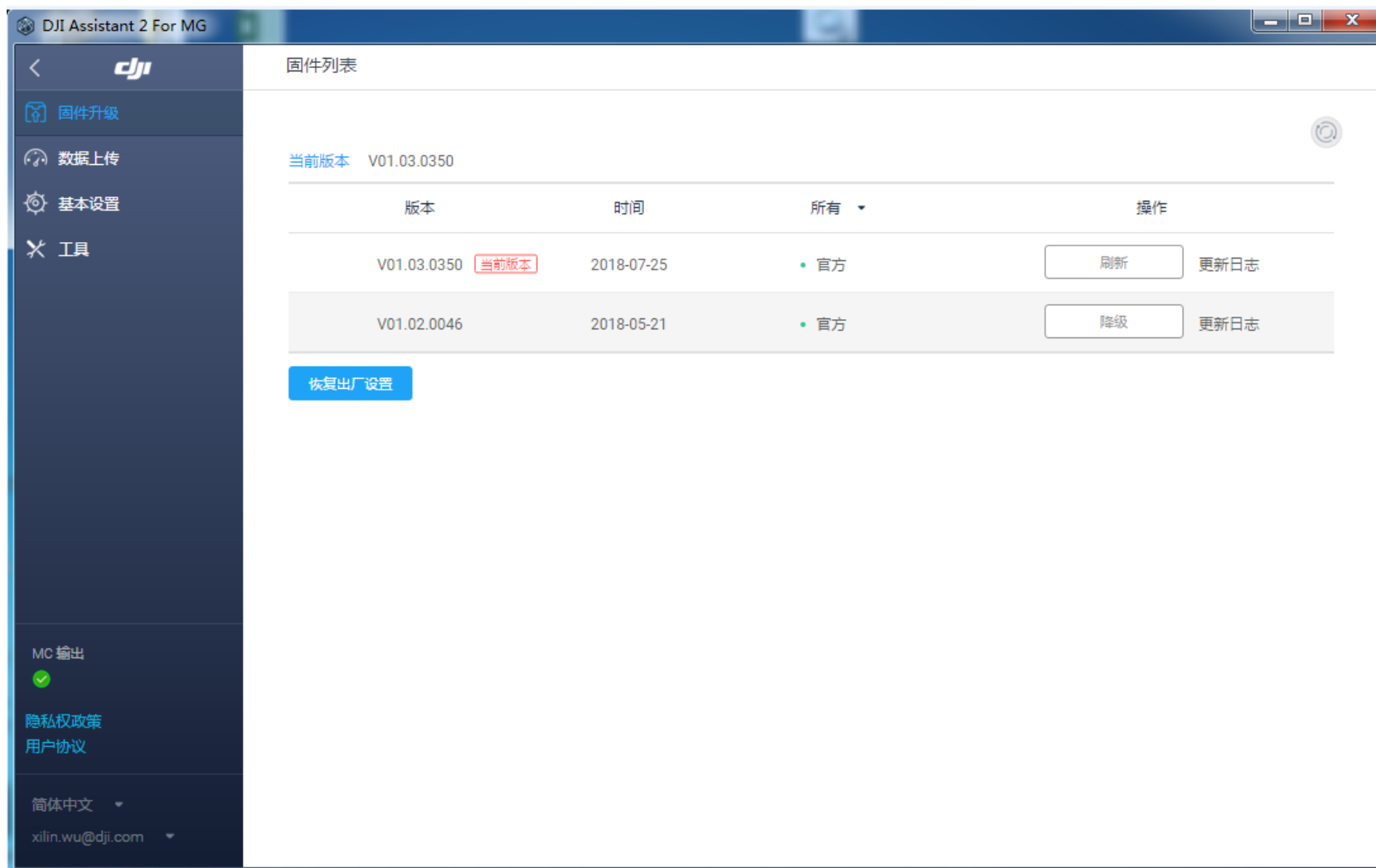
For more details about data analysis, refer to *Flight Controller Data Analysis Series Tutorials*.

Functional Inspection



Click “Basic Settings” and “ESC Settings” and check the rotation direction of the motors 1 to 8 one by one. Motor 1, motor 3, motor 5, and motor 7 rotate counterclockwise while motor 2, motor 4, motor 6, and motor 8 rotate clockwise.

Functional Inspection



Click “Firmware Update” and “Firmware List” and check whether the MG-1P needs to be updated. Click “Release Note” to see the firmware’s update log.

Checking the Remote Controller



Checking the Remote Controller

1. Check whether the remote controller's upper and lower shells are broken, or the antennas are cracked.
Push the control sticks and check the power button and dials, and check whether they can work normally.
2. Power on the remote controller, connect to the internet and enter the operation screen. Press the link button on the remote controller and link the remote controller to the aircraft. Press the button on the link board (mounted to the central board) on the aircraft and the indicator will be solid green. If the indicator does not light up after the button is pressed, the button is damaged.

The remote controller cannot link with the drone.

- ① Check whether the drone's firmware version matches the remote controller's version.
- ② Disassemble the lower shell, check whether the flat cable connecting the SDR module and central board and the cable connecting the central board and link board are damaged.
- ③ Replace the link board, SDR module, and central board, then perform the linking test.
- ④ Disassemble the remote controller and check whether the antenna is loose.
- ⑤ Check whether the flat cables are connected firmly or damaged.
- ⑥ Replace the main board and test again.



Checking the Remote Controller



The aircraft status, remote controller status, and app version can be checked in the general settings. After linking, the aircraft status and warnings can be displayed.

Checking the Remote Controller



2. The remote controller cannot be powered on.

Solution: Charge the remote controller and check whether the remote controller's battery level is enough to power on.

If yes, but the issue persists, replace the remote controller's power button.

3. The remote controller's screen cannot be turned on.

Solution:

- ① Disassemble the remote controller and check whether the screen's flat cable is connected correctly or damaged.
- ② Replace the screen and test again.

4. The remote controller's

Solution: After it is confirmed that the buttons and cables are normally functioning, restore the factory defaults and update the remote controller's firmware version.

1. The drone's firmware cannot be updated.

Solution:

- ① Check whether the USB cable connecting the SDR module and the central board is connected firmly or damaged.
- ② Check whether the USB port on the central board is damaged.

2. After the drone is connected to the PC, the USB port cannot be identified or DJI Assistant 2 for MG does not work.

Solution: Check whether the cable connecting the USB port on the central board and main controller, the cable connecting the USB port on the central board and SDR, and the cable connecting the SDR and RF are connected firmly or damaged.

If not, use the test SDR module and test again.

3. The remote controller prompts that there is an error with the navigation system.

Solution: Use the test GPS module and test again. If the issue persists, replace the main controller.

Main Controller Communication Error (DJI MG App Warning Report)



1. A “IMU Error 10” warning is shown in the app.

Checking method: Restart-Calibration-Replace the GPS-Replace the Main Controller

Solution: Restart the product and place the aircraft on level ground for a while. If it is unable to pass the initial attitude test within 30s, replace the flight controller.

2. The app shows that there is an error with the ESC.

The motor's indicator blinks red and yellow alternatively.

Checking method: ESC-Y-type Cable-Motor-Main Controller-LED

3. There is an error with the communication between the Intelligent Battery and the main controller

Checking method: Battery-Aircraft's Socket check whether the central board's cable is loose or broken. Check the cable connecting the central board and GPS module. Replace the power distribution board and smart socket.

Magnetic Interference (DJI MG App Warning Report)



1. The app reports that there is a compass error during takeoff.

Checking method: Fly the drone in a place free of interference – Calibrate the compass – Check whether the compass is mounted correctly – Check the landing gear's cable – GPS.

2. The app reports that the IMU data is too large.

Checking method: Calibrate the IMU in different directions – Restart the product – Replace the main controller.

3. The Home Point is not recorded when the drone takes off.

Checking method: Check the flying environment and weather – Check the distance between the aircraft and remote controller

4. The app shows that there is a strong magnetic field during operation.

Checking method: Check the linear distance between the aircraft and the user - Check whether the drone has interference (the drone may be interfered with by the antenna masts and high voltage lines). – Check the remote controller's battery level and GPS signal.

ESC Throttle Cable Error (DJI MG App Warning Report)



1. The ESC indicator blinks red and yellow alternatively.

Checking method: ESC – Y-type Cable – Motor – 485 Adaptive Board – Main Controller – Motor's LED.

2. The drone was water damaged, and the app prompts that there is an error with the ESC or throttle cable.

Checking method: ESC – Y-type Cable – 485 Adaptive Board – Motor – Motor's LED – Central Board – Power Distribution Board – Main Controller.

3. The firmware update is failed with DJI Assistant 2 for MG, and it shows that there is an error with the ESC.

Checking method: Update the firmware again – Check the ESC – Motor – 485 Adaptive Board – Y-type Cable – Motor.



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**THANK
YOU**

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